

THE CURIOSITY CUP 2025

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Trends in Expenditures and Associated Factors for Diabetes Patients Among the U.S. Civilian Noninstitutionalized Population, 2012–2022

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ABSTRACT

Background: The number of diabetes patients continues to grow, and healthcare expenditures have been rising yearly, impacting the U.S. healthcare system. This study aims to analyze recent trends in overall medical expenses for diabetes patients and identify key determinants of rising costs to provide empirical evidence for policymakers.

Methods: Using data from the Medical Expenditure Panel Survey (MEPS) from 2012 to 2022, we apply regression modeling to examine the effects of demographic, socioeconomic, and health-related factors on healthcare expenditures.

Results: The findings indicate a general increase in diabetes-related healthcare expenditures, with significant differences across insurance types and health conditions. Key influencing factors include provider type, cardiovascular disease, health status, race/ethnicity, and insulin use.

Conclusion: This study provides a comprehensive analysis of medical expenses for diabetes patients, identifying key cost drivers. The findings offer valuable insights for future healthcare policy formulation to reduce economic burdens and optimize diabetes management strategies.

INTRODUCTION

Diabetes affects over 38 million Americans, driving rising healthcare costs (CDC, 2024). Over the past decade, U.S. diabetes-related expenditures surged from \$227 billion (2012) to \$307 billion (2022) (ADA, 2023), reflecting the high costs of managing complications like hospitalizations, medications, and conditions such as heart disease and amputations, which account for 48%- 64% of lifetime healthcare costs.

The CDC reports 133 million Americans now have diabetes or prediabetes, with many cases undiagnosed, increasing costs due to delayed care (ADA, 2023). Despite this growing burden, the last major expenditure analysis covered 2002-2011 (Zhou et al., 2013). This study fills a critical gap by analyzing 2012-2022 trends, providing updated insights to inform healthcare strategies.

PROBLEM STATEMENT

The problem addressed in this paper is the escalating healthcare costs associated with the rising prevalence of diabetes in the United States, which remain insufficiently analyzed in recent years (Egede, 2015). Utilizing Medical Expenditure Panel Survey (MEPS) data from 2012 to 2022, this study aims to update and deepen our understanding of the expenditure trends and to elucidate the demographic, socioeconomic, and health-related drivers of these

costs. Specifically, we focus on diabetes patients within the MEPS dataset, applying sample weights to ensure generalizability and employing MEPS's comprehensive data structure to conduct a detailed individual-level analysis. The use of statistical techniques like annual expenditure summaries, regression modeling, and subgroup comparisons is intended to identify and confirm the key factors influencing healthcare expenditures. By analyzing cost trends, this study provides insights for future healthcare strategies, helping mitigate diabetes-related financial and health burdens in the U.S.

EXPLORATORY DATA ANALYSIS

Our dataset includes 1,420 variables from 22,431 individuals across multiple survey rounds. Total Expenditures (TOTEXP) is the dependent variable, with the goal of identifying key cost drivers.

Independent variables fall into three categories:

- **Binary** (e.g., disease presence: "yes"/"no").
- **Numeric** (e.g., AGELAST for age).
- **Nominal** (e.g., MARRY for marital status).

Initial analysis revealed outliers and zero values in TOTEXP, indicating some individuals reported no healthcare expenses. Additionally, categorical and binary data contain values like -15, -8, -7, and -1, representing missing or inapplicable responses. These data issues will be addressed in the next analysis step.

DATA CLEANING AND PROCESSING

After Exploratory Data Analysis, we implemented systematic data cleaning to ensure accuracy and reliability. Key steps included selecting diabetic patients, retaining relevant variables, and handling missing values and outliers.

1. Selecting Diabetic Patients

We filtered the dataset to include only individuals diagnosed with diabetes (DIABDX_M18 = 1) to maintain study relevance.

2. Variable Selection

Retained 39 key variables across demographics, comorbidities, diabetes management, expenditures, and weighting factors.

3. Handling Missing Values and Outliers

- Missing Values: Recoded -15, -8, -7, -1 as missing to prevent bias.
- Outliers: Used the IQR method to remove extreme values in TOTEXP ($Q1 - 3 \times IQR$ or $Q3 + 3 \times IQR$).
- Zero Expenditure Records: Removed TOTEXP = 0 cases to ensure meaningful expenditure insights.

DATA MODEL AND ANALYSIS

In this section, we present our analysis in two parts. The first part examines expenditure trends from 2012 to 2022, highlighting overall patterns and significant shifts in healthcare spending for diabetes patients. The second part focuses on 2018-2022, analyzing key factors driving expenditures, including demographic, socioeconomic, and health status variables that contribute to cost variations.

FIRST MAIN TOPIC <TREND IN EXPENDITURES FROM 2012 TO 2022>

1. Adjustment to 2022 Dollars

- Total Expenditures were adjusted for inflation using the Consumer Price Index (CPI) to ensure comparability across years.

2. Trend Analysis of Adjusted Expenditures

- **Mean Calculation (PROC SURVEYMEANS):** Computed mean Adjusted Total Expenditures while accounting for survey design.
- **Visualization (PROC SGPLOT):** Plotted mean expenditures (2012–2022) to illustrate spending trends.

3. Expenditures by Insurance Type

- **Classification (INSCOV):** Individuals were classified into Private Insurance (with any private coverage), Public Only (covered solely by public programs), and Uninsured (without any health insurance).
- **Mean Calculation (PROC SURVEYMEANS):** Computed expenditures by insurance type.
- **Visualization (PROC SGPLOT):** Created a line chart to compare expenditure trends across insurance groups from 2012 to 2022 with distinct colors and a legend.

SECOND MAIN TOPIC <IDENTIFY KEY FACTORS THAT INFLUENCE EXPENDITURES>

1. Expenditure Distribution and Statistical Significance

- **Distribution Analysis (PROC SURVEYFREQ):** Assessed sample representativeness across demographics, health conditions, and healthcare access.
- **Mean Comparison (PROC SURVEYMEANS):** Calculated weighted means and standard errors for TOTEXP across variable groups.
- **Significance Testing (PROC SURVEYREG):** Used regression analysis to determine if expenditure differences were statistically significant (F-test).

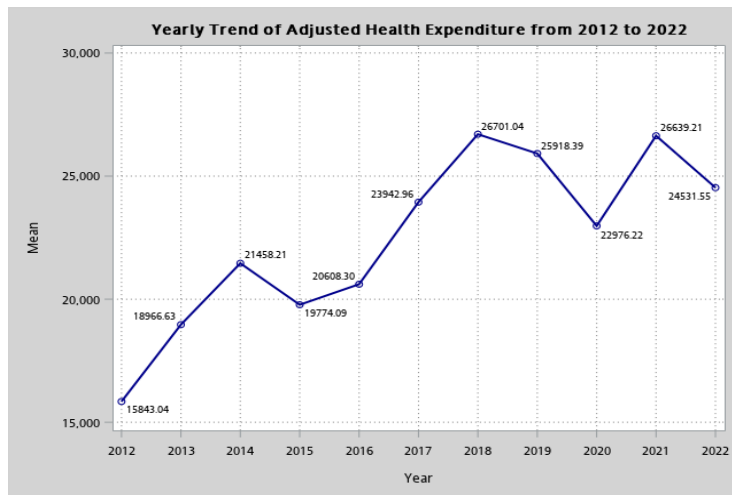
2. Multiple Regression Analysis

- **Model 1 (Manually Selected, PROC SURVEYREG):** Identified significant variables annually, capturing year-specific predictors and trends.
- **Model 2 (Automated Selection, %StepSvyreg):** Implemented stepwise selection to optimize variable inclusion while accounting for survey design. This method is based on the framework from "SAS® Model Selection Macros for Complex Survey Data Using PROC SURVEYLOGISTIC/SURVEYREG" by Fang Wang & Hee-Choon Shin (NORC at the University of Chicago).

Combining both methods ensures comprehensive analysis—Method 1 tracks yearly trends, while Method 2 enhances selection efficiency, balancing interpretability and statistical robustness.

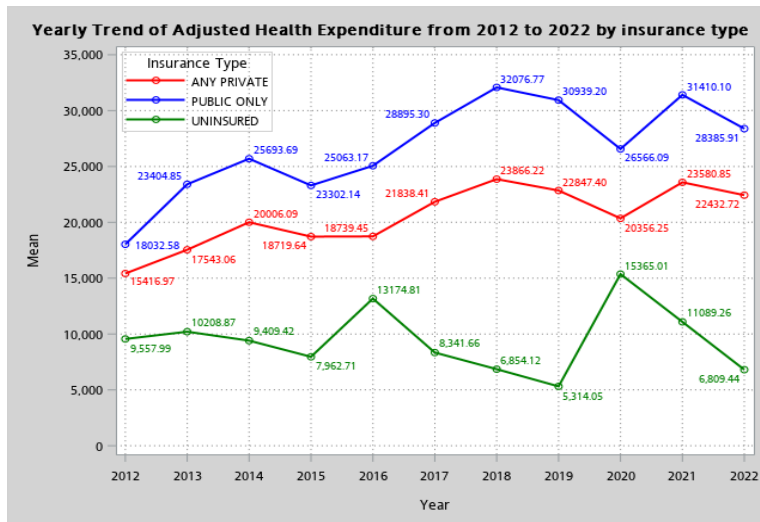
RESULTS

- **Trend in Expenditures from 2012 to 2022**



The trend of adjusted health expenditure from 2012 to 2022 shows an overall increase, rising from 15,843.04 in 2012 to 24,531.55 in 2022. Expenditures increased steadily from 15,843.04 in 2012 to 21,458.21 in 2014, followed by a decline to 19,774.09 in 2015. After a rise to 26,701.04 in 2018, expenditures slightly decreased to 25,918.39 in 2019 and further dropped to 22,976.22 in 2020. In 2021, expenditures rebounded to 26,639.21 before decreasing again to 24,531.55 in 2022.

Trend in Expenditures from 2012 to 2022 by Insurance Type



1) Private Insurance:

The expenditure was 15,416.57 in 2012 and showed an increasing trend, reaching 22,472.72 in 2022. The highest expenditure was recorded in 2018 (23,866.22) and 2019 (22,847.40).

A drop occurred in 2020 (20,365.25) before rising again in 2021 (23,805.85) and slightly dropping again in 2022 (22,472.72).

2) Public Insurance:

The expenditure was 18,032.58 in 2012 and showed an increasing trend, increasing to 25,693.69 in 2014. After a drop in 2015, it shows a growing trend again and the highest recorded expenditure was in 2018 (32,076.77), followed by a slight decrease in 2019 (30,939.20). A drop was observed in 2020 (25,656.09) before recovering in 2021 (31,410.10) and dropping again in 2022 at 28,385.91.

3) Uninsured Individuals:

The lowest expenditure was consistently observed for uninsured individuals, starting at 9,557.99 in 2012 to 7,942.71 in 2015. A peak occurred in 2016 (13,174.81), but a sharp decline followed in 2019 (5,314.05). The lowest value recorded was in 2019 (5,314.05), with some recovery in 2020 (13,565.01) and a drop again in 2021 (11,089.26) and 2022 (6,809.44).

- **Identify Key Factors That Influence Expenditures**

2022				
Obs	Label1	cValue1	nValue1	
1	R-Square	0.245	0.245008	
2	Root MSE	26767	25767	
3	Denominator DF	231	231.000000	
Obs Effect FValue ProbF				
1	Model	1747781.173	<.0001	
2	Intercept	21.38	<.0001	
3	TYPEPE42	9.47	<.0001	
4	TMTKUS42	2.97	0.0001	
5	STRKDX	4.48	0.0353	
6	RTHLTH53	5.50	<.0001	
7	RACETHX	5.48	0.0019	
8	POVCAT	3.22	0.0135	
9	OHRTDX	17.12	<.0001	
10	MMHLTH53	1.45	0.2283	
11	MARRY	1.68	0.1958	
12	INSCOV	5.29	0.0057	
13	HAVEUS42	1.93	0.1478	
14	EMPHDX	0.63	0.4274	
15	ADLHLP31	3.64	0.0279	
16	AGEGROUP	0.36	0.5477	
17	ARTHDX	15.95	<.0001	
18	ASTHDX	3.96	0.0163	
19	CANCERDX	2.33	0.0996	
20	CHIRON31	0.99	0.3719	
21	CHOLDX	22.68	<.0001	
22	DSCHNV53	3.08	0.0479	
23	DSDIET53	2.40	0.0933	
24	DSSENV53	0.88	0.4926	
25	DSSENV53	3.44	0.0336	
26	DSFLNV53	3.27	0.0398	
27	DSFTNV53	0.71	0.4946	
28	DSINSU53	4.14	0.0172	
29	DSKDIN53	2.16	0.1171	

2021				
Obs	Label1	cValue1	nValue1	
1	R-Square	0.1937	0.193646	
2	Root MSE	27702	27702	
3	Denominator DF	252	252.000000	
Obs Effect FValue ProbF				
1	Model	16	<.0001	
2	Intercept	35.80	<.0001	
3	TYPEPE42	2.66	0.0004	
4	STRKDX	1.01	0.3107	
5	RTHLTH53	10.88	<.0001	
6	RACETHX	6.48	<.0001	
7	POVCAT	4.03	0.0024	
8	OHRTDX	9.08	0.0002	
9	MMHLTH53	2.49	0.0319	
10	MARRY	1.67	0.1417	
11	INSCOV	1.52	0.2208	
12	HAVEUS42	0.52	0.5997	
13	EMPHDX	2.11	0.1474	
14	TMTKUS42	1.44	0.0891	
15	AGEGROUP	2.84	0.0932	
16	ARTHDX	2.84	0.0932	
17	ASTHDX	3.24	0.0725	
18	CANCERDX	5.98	<.0001	
19	CHOLDX	0.59	0.5567	
20	DSDIET53	1.98	0.1402	
21	DSSENV53	2.78	0.0623	
22	DSSENV53	0.40	0.5704	
23	DSFLNV53	2.56	0.0794	
24	DSFTNV53	1.43	0.2317	
25	DSINSU53	10.95	<.0001	
26	DSKDIN53	0.58	0.5593	

2020				
Obs	Label1	cValue1	nValue1	
1	R-Square	0.1714	0.171450	
2	Root MSE	23399	23399	
3	Denominator DF	171	171.000000	
Obs Effect FValue ProbF				
1	Model	52	<.0001	
2	Intercept	50.97	<.0001	
3	TYPEPE42	3.29	<.0001	
4	TMTKUS42	5.23	<.0001	
5	STRKDX	2.80	0.0960	
6	RTHLTH53	13.02	<.0001	
7	RACETHX	1.64	0.0071	
8	OHRTDX	18.28	<.0001	
9	MMHLTH53	3.27	0.0076	
10	INSCOV	0.77	0.4827	
11	EMPHDX	0.07	0.8814	
12	ADLHLP31	0.82	0.4403	
13	AGEGROUP	1.51	0.2202	
14	ARTHDX	2.48	0.1185	
15	ASTHDX	9.05	0.0031	
16	CANCERDX	27.94	<.0001	
17	CHOLDX	3.98	0.0478	
18	DSCHNV53	5.15	0.0004	
19	DSSENV53	1.95	0.1448	
20	DSSENV53	2.35	0.0984	
21	DSFLNV53	0.85	0.4203	
22	DSFTNV53	0.03	0.9508	
23	DSINSU53	10.51	<.0001	
24	DSKDIN53	5.77	0.0037	

2019				
Obs	Label1	cValue1	nValue1	
1	R-Square	0.2113	0.211356	
2	Root MSE	25381	25381	
3	Denominator DF	174	174.000000	
Obs Effect FValue ProbF				
1	Model	1892	<.0001	
2	Intercept	1.28	0.2599	
3	TYPEPE42	2.05	0.0135	
4	TMTKUS42	4.23	0.0001	
5	STRKDX	5.53	0.0198	
6	RTHLTH53	11.74	<.0001	
7	RACETHX	1.25	0.0134	
8	POVCAT	1.45	0.2205	
9	OHRTDX	4.34	0.0145	
10	MMHLTH53	1.59	0.1651	
11	MARRY	1.49	0.2088	
12	INSCOV	22.52	<.0001	
13	HAVEUS42	3.97	0.0206	
14	EMPHDX	0.06	0.8365	
15	ADLHLP31	0.09	0.0002	
16	AGEGROUP	2.39	0.1241	
17	ARTHDX	1.13	0.2940	
18	ASTHDX	1.48	0.2260	
19	CANCERDX	55.88	<.0001	
20	CHOLDX	1.67	0.1921	
21	DSCHNV53	1.09	0.3185	
22	DSSENV53	3.47	0.0332	
23	DSFLNV53	0.91	0.4044	
24	DSFTNV53	25.67	<.0001	
25	DSINSU53	12.91	<.0001	
26	DSKDIN53	1.02	0.3638	

2018				
Obs	Label1	cValue1	nValue1	
1	R-Square	0.228	0.218011	
2	Root MSE	25040	25040	
3	Denominator DF	140	140.000000	
Obs Effect FValue ProbF				
1	Model	112	<.0001	
2	Intercept	35.43	<.0001	
3	TYPEPE42	8.84	<.0001	
4	STRKDX	13.21	0.0004	
5	SEX	5.60	0.0193	
6	RTHLTH53	4.35	0.0010	
7	RACETHX	12.08	<.0001	
8	POVCAT	2.32	0.0595	
9	OHRTDX	25.65	<.0001	
10	MMHLTH53	0.37	0.4706	
11	MARRY	3.53	0.0049	
12	INSCOV	20.64	<.0001	
13	HAVEUS42	0.54	0.5865	
14	EMPHDX	0.27	0.6060	
15	DSMED53	3.05	0.0286	
16	ADLHLP31	1.86	0.1589	
17	AGEGROUP	0.10	0.7540	
18	ARTHDX	45.37	<.0001	
19	ASTHDX	2.98	0.0864	
20	CANCERDX	0.72	0.4877	
21	CHIRON31	0.94	0.3931	
22	CHOLDX	2.41	0.0933	
23	DSCHNV53	3.61	0.0028	
24	DSSENV53	0.51	0.4883	
25	DSSENV53	0.95	0.3770	
26	DSFLNV53	0.34	0.7094	
27	DSFTNV53	0.79	0.4559	
28	DSINSU53	6.08	0.0029	
29	DSKDIN53	6.50	0.0020	

After univariate regression, we built Model 1 using multiple regression, with R² values ranging from 17.14% to 24.5% (highest in 2022: 0.2450, lowest in 2020: 0.1714). Key predictors consistently significant across all years included TYPEPE42 (Provider Type), STRKDX (Stroke Diagnosis), RTHLTH53 (Health Status), RACETHX (Race/Ethnicity), OHRTDX (Heart Disease), and DSINSU53 (Insulin Treatment).

2022				
Obs	Label1	cValue1	nValue1	
1	R-Square	0.2192	0.219164	
2	Root MSE	27050	27050	
3	Denominator DF	231	231.000000	
Obs Effect FValue ProbF				
1	Model	40	<.0001	
2	Intercept	74.33	<.0001	
3	TYPEPE42	19.17	<.0001	
4	ARTHDX	21.35	<.0001	
5	OHRTDX	20.94	<.0001	
6	RTHLTH53	8.33	<.0001	
7	RACETHX	6.94	<.0001	
8	CHOLDX	9.18	0.0001	
9	ASTHDX	11.12	0.0010	
10	INSCOV	6.85	0.0013	
11	TMTKUS42	3.75	0.0014	
12	DSINSU53	5.89	0.0032	
13	POVCAT	3.46	0.0001	
14	STRKDX	3.98	0.0200	
15	DSFLNV53	4.17	0.0167	
16	DSSENV53	3.97	0.0203	
17	ADLHLP31	3.90	0.0216	

2021				
Obs	Label1	cValue1	nValue1	
1	R-Square	0.1741	0.174143	
2	Root MSE	27928	27928	
3	Denominator DF	252	252.000000	
Obs Effect FValue ProbF				
1	Model	21	<.0001	
2	Intercept	7.84	0.0053	
3	CANCERDX	10.97	<.0001	
4	INSCOV	3.33	0.0373	
5	RTHLTH53	12.20	<.0001	
6	DSINSU53	12.58	<.0001	
7	ADLHLP31	2.60	0.0760	
8	TMTKUS42	0.54	0.7760	
9	OHRTDX	22.92	<.0001	
10	TYPEPE42	4.35	<.0001	
11	RACETHX	7.51	<.0001	
12	HAVEUS42	0.81	0.4438	
13	STRKDX	1.01	0.3127	
14	DSSENV53	0.63	0.5332	
15	CHIRON31	0.12	0.8873	
16	AGEGROUP	3.53	0.0615	

2020				
Obs	Label1	cValue1	nValue1	
1	R-Square	0.1517	0.151721	
2	Root MSE	23992	23992	
3	Denominator DF	171	171.000000	
Obs Effect FValue ProbF				
1	Model	80	<.0001	
2	Intercept	103.63	<.0001	
3	RTHLTH53	17.89	<.0001	
4	CANCERDX	27.72	<.0001	
5	TYPEPE42	7.22	<.0001	
6	OHRTDX	13.63	0.0003	
7	DSINSU53	16.02	<.0001	
8	TMTKUS42	5.35	<.0001	
9	RACETHX	4.40	0.0021	
10	ASTHDX	8.87	0.0033	
11	MMHLTH53	4.83	0.0004	
12	AGEGROUP	8.05	0.0051	
13	CHOLDX	6.17	0.0140	
14	EMPHDX	4.17	0.0427	

2019				
Obs	Label1	cValue1	nValue1	
1	R-Square	0.1971	0.197077	
2	Root MSE	25467	25467	
3	Denominator DF	174	174.000000	
Obs Effect FValue ProbF				
1	Model	474	<.0001	
2	Intercept	2.90	0.0905	
3	CANCERDX	170.95	<.0001	
4	INSCOV	35.11	<.0001	
5	RTHLTH53	14.85	<.0001	
6	DSINSU53	15.72	<.0001	
7	ADLHLP31	9.07	0.0002	
8	TMTKUS42	4.64	0.0002	
9	OHRTDX	10.94	0.0012	
10	TYPEPE42	2.43	0.0020	
11	RACETHX	3.71	0.0064	
12	HAVEUS42	4.42	0.0134	
13	STRKDX	6.01	0.0152	

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APPENDIX

Code Sample:

/ 1. Analysis of Healthcare Expenditure Distribution and Statistical Significance*/*

%macro var(var);

%do y=**2018** **%to** **2022**;

proc surveyfreq data=tmp.dm3_&y.missing;

strata VARSTR;

cluster VARPSU;

weight PERWTF;

tables &var.;

run;

proc surveymeans data=tmp.dm3_&y.missing;

strata VARSTR;

cluster VARPSU;

weight PERWTF;

class &var.;

var TOTEXP;

domain &var.;

run;

proc surveyreg data=tmp.dm3_&y.missing;

strata VARSTR;

cluster VARPSU;

weight PERWTF;

class &var.;

model TOTEXP = &var. / solution;

run;

%end;

%mend;

%var(AGEGROUP);

*/*2. Multiple Regression Analysis*/*

*/*methods 1*/*

*/*2022*/*

proc surveyreg data=tmp.dm3_2022 missing;

strata VARSTR;

cluster VARPSU;

class TYPEPE42 TMTKUS42 STRKDX RTHLTH53 RACETHX POVCAT OHRTDX MNHLTH53 MARRY
INSCOV HAVEUS42 EMPHDX ADLHLP31 Agegroup ARTHDX ASTHDX CANCERDX CHBRON31

```

CHOLDX DSCHNV53 DSDIET53 DSEYNV53 DSEYPR53 DSFLNV53 DSFTNV53 DSINSU53
DSKIDN53;
model TOTEXP=TYPEPE42 TMTKUS42 STRKDX RTHLTH53 RACETHX POVCAT OHRTDX
MNHLTH53 MARRY INSCOV HAVEUS42 EMPHDX ADLHLP31 Agegroup ARTHDX ASTHDX
CANCERDX CHBRON31 CHOLDX DSCHNV53 DSDIET53 DSEYNV53 DSEYPR53 DSFLNV53
DSFTNV53 DSINSU53 DSKIDN53/solution;
weight PERWTF;
run;
/*methods 2-stepwise-reference: https://www.norc.org/content/dam/norc-org/pdfs/MWSUG-2011-SA02.pdf/
/*2022*/
%let input=tmp. dm3_2018;
%let ind=AGEGROUP SEX RACETHX MARRY POVCAT CANCERDX OHRTDX STRKDX EMPHDX
CHOLDX ARTHDX ASTHDX CHBRON31 RTHLTH53 MNHLTH53 ADLHLP31 HAVEUS42 TMTKUS42
TYPEPE42 INSCOV DSFTNV53 DSEYNV53 DSCHNV53 DSFLNV53 DSKIDN53 DSEYPR53 DSDIET53
DSINSU53 DSMED53 DSCONF53;
%let classv=AGEGROUP SEX RACETHX MARRY POVCAT CANCERDX OHRTDX STRKDX EMPHDX
CHOLDX ARTHDX ASTHDX CHBRON31 RTHLTH53 MNHLTH53 ADLHLP31 HAVEUS42 TMTKUS42
TYPEPE42 INSCOV DSFTNV53 DSEYNV53 DSCHNV53 DSFLNV53 DSKIDN53 DSEYPR53 DSDIET53
DSINSU53 DSMED53 DSCONF53;
%let strata=VARSTR;
%let cluster=VARPSU;
%let weight=PERWTF;
%let dep=TOTEXP;
%StepSvyreg(
    &dep.,
    &ind.,
    &classv.,
    &input.,
    0.05,
    0.05,
    &weight.,
    &strata.,
    &cluster.,
    3    );
ods trace on;
ods trace off;

```